SIMATS ENGINEERING ASSESSMENT
TOOL 2

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Marks:88

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Data Interpretation (Medium)

Weightage: 20%

QUESTIONS: **Total Marks: 20**

1: HTTP Response Analysis

A web server returns the following HTTP response header:

```
HTTP/1.1 200 OK
Date: Mon, 01 Apr 2026 10:00:00 GMT
Content-Type: text/html
Content-Length: -150
Connection: keep-alive
```

Questions:

1. Identify any incorrect or invalid fields in the header.
2. Analyze the issue with `Content-Length` and explain its impact.
3. Determine whether the status code matches the response correctness.
4. Suggest corrections for the identified errors.

2: **DOM Structure Analysis**

Consider the following DOM tree representation:

```
<html>
  <head>
  <title>TestPage</title>
</head>
  <body>
  <div>
  <p>HelloWorld
  </div>
 </body>
 </html>
```

Questions:

1. Identify missing or improperly closed elements.
2. Analyze how the browser will interpret this structure.
3. Determine the impact on rendering and layout.
4. Suggest corrections to make the DOM valid.

3: GET vs POST in Login System

A login system uses two different methods:

Method	URL Example	Data Visibility	Security Level
GET	/login?user=admin&pass=1234	Visible in URL	Low
POST	/login	Hidden in body	Moderate

Questions:

1. Interpret the differences in data transmission between GET and POST.
2. Analyze which method is more suitable for login and why.
3. Identify security risks in using GET for authentication.
4. Suggest best practices for secure login implementation.

4: HTML Code Inspection

Analyze the following HTML snippet:

```
<!DOCTYPE html>
<html>
<head>
<title>Sample</title>
</head>
<body>
<h1>Welcome
<p>This is a paragraph
<img src="image.jpg">
</body>
</html>
```

Questions:

1. Identify missing closing tags.
2. Analyze issues with the `<img>` tag.
3. Determine how browsers will handle these errors.
4. Suggest a corrected version of the code.

5: Browser-Server Communication Flow

The following sequence describes a web request:

1. User enters a URL in the browser
2. DNS lookup occurs
3. HTTP request is sent
4. Server processes request
5. Response is returned
6. Browser renders page

Questions:

1. Interpret each step in the communication flow.
2. Identify where delays can occur.
3. Analyze the role of DNS in the process.
4. Suggest optimizations to improve performance. <!DOCTYPE html>

```
<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>CSA43 - Data Interpretation Assignment</title>

  <style>       body { font-family: Arial, sans-serif; line-height: 1.6; margin: 40px;

color: #333; }       h1 { text-align: center; color: #0056b3; }       h2 { color: #d9534f;

border-bottom: 2px solid #eee; padding-bottom: 5px; }       h3 { color: #0275d8; }
```

pre { background: #f4f4f4; padding: 15px; border-radius: 5px; overflow-x: auto; }

code { font-family: Consolas, monospace; color: #c7254e; }

.question-block { margin-bottom: 30px; padding: 20px; border: 1px solid #ddd; border-radius: 5px; }

</style>

</head>

<body>

<h1>Internet Programming Assignment (CSA43)</h1>

Course Outcome: CO-1[cite: 1]

<h2>Question 1: HTTP Response Analysis</h2>

1. Incorrect/invalid fields: The day of the week in the <code>Date</code> header may not match the actual calendar date for April 1, 2026 (it is a Wednesday, not Monday)[cite: 1]. Also, it lacks standard headers like <code>Server</code>.
2. Content-Length issue: If the <code>Content-Length: 150</code> does not exactly match the byte size of the returned body, the browser will either truncate the page (if too small) or hang waiting for more data (if too large)[cite: 1].

3. Status code: <code>200 OK</code> is the standard response for successful HTTP requests, meaning the response correctness matches a successful delivery[cite: 1].

4. Suggested corrections: Ensure the date format strictly matches the calendar day and ensure the <code>Content-Length</code> is dynamically calculated to match the exact byte size of the payload[cite: 1].

<h2>Question 2: DOM Structure Analysis</h2>

1. Missing/improperly closed elements: The <code><p></code> tag is opened but never closed before the closing <code></div></code>[cite: 2].

2. Browser interpretation: Browsers are forgiving and will automatically infer the closing <code></p></code> tag before the <code></div></code> closes[cite: 2].

3. Impact on rendering: While modern browsers will fix it, relying on error-correction can cause unpredictable layout shifts or CSS styling errors in complex documents[cite: 2].

4. Suggested corrections:

<pre><code><html>

<head>

  <title>TestPage</title>

</head>

<body>

  <div>

    <p>HelloWorld</p>

  </div>

</body>

</html></code></pre>

    <h2>Question 3: GET vs POST in Login System</h2>

    1. Data transmission differences: GET transmits data appended to the URL query string. POST transmits data hidden within the HTTP request body[cite: 2].

    2. Suitability for login: POST is more suitable because it does not expose sensitive credentials in the URL[cite: 2].

    3. Security risks of GET: Passwords will be visible in the URL, browser history, server access logs, and can be easily intercepted or shoulder-surfed[cite: 2].

    4. Best practices: Always use the POST method for authentication over a secure HTTPS encrypted connection[cite: 2].

    <h2>Question 4: HTML Code Inspection</h2>

    1. Missing closing tags: Both the <code><h1></code> and <code><p></code> tags are missing their respective closing tags[cite: 3].

2. Issues with <img> tag: The image tag is missing the required <code>alt</code> attribute for accessibility and valid HTML standards[cite: 3].

3. Browser handling: The browser will likely treat the subsequent elements (like the paragraph and image) as part of the <code><h1></code> block if not autocorrected, leading to massive text rendering[cite: 3].

4. Corrected Version:

<pre><code><!DOCTYPE html>

<html>

<head>

  <title>Sample</title>

</head>

<body>

  <h1>Welcome</h1>

  <p>This is a paragraph</p>

  <img src="image.jpg" alt="Sample Image Description">

</body>

</html></code></pre>

    <h2>Question 5: Browser-Server Communication Flow</h2>

    1. Interpretation of steps: The user requests a resource (1), the domain is translated to an IP (2), the browser sends data to the server (3), the server generates the response (4), sends it back (5), and the browser paints it on screen (6)[cite: 3].

    2. Where delays can occur: DNS lookup (Step 2), network latency in requests/responses (Steps 3 and 5), heavy backend database queries (Step 4), and large image/DOM rendering (Step 6)[cite: 3].

    3. Role of DNS: It acts as the internet's phonebook, translating humanreadable URLs into machine-readable IP addresses[cite: 3].

    4. Optimizations: Implement browser caching, use Content Delivery Networks (CDNs), minify HTML/CSS/JS files, and use DNS prefetching[cite: 3].

</body>

</html>

Internet Programming Assignment Solutions

Question 1: HTTP Response Analysis

1. Incorrect/invalid fields: The `Date` header day (Mon) does not match April 1, 2026 (Wednesday). More importantly, the `Content-Length` is negative (`-150`), which is invalid as length must be a non-negative integer.

2. Content-Length issue and impact: A negative `Content-Length` violates HTTP protocols. The browser or proxy will likely reject the response, drop the connection, or fail to parse the body, resulting in a blank page or network error.

3. Status code correctness: `200 OK` indicates success, but this contradicts the severely malformed headers (invalid length). The response is technically corrupt despite the 200 status.

4. Suggested corrections: Fix the date to exactly match the calendar (`Wed, 01 Apr 2026`) and correct the `Content-Length` to the exact positive byte size of the payload (e.g., `150`).

Question 3: GET vs POST in Login System

1. Differences in data transmission: GET appends form data to the URL query string. POST places form data inside the HTTP request body.

2. Method suitable for login: POST is required for logins because it keeps sensitive data out of the URL, browser history, and server logs.

3. Security risks of GET: Sending credentials via GET exposes passwords in plain sight within the browser address bar, shoulder-surfing, browser history, bookmarks, and server access logs.

4. Best practices for secure login: Use the POST method, enforce HTTPS (SSL/TLS) for encrypted transmission, implement rate-limiting, and hash passwords on the server.

Question 2: DOM Structure Analysis

1. Missing/improperly closed elements: The `<p>` tag is opened but not closed before the parent `</div>` closes.

2. Browser interpretation: Modern browsers use error-handling to auto-close the `<p>` tag just before the `</div>` is closed.

3. Impact on rendering and layout: While usually visually corrected by the browser, relying on parser error-correction can cause unpredictable layout shifts, CSS styling issues, or JavaScript DOM traversal bugs.

4. Suggested corrections:

```
<html>
<head>
    <title>TestPage</title>
</head>
<body>
    <div>
        <p>HelloWorld</p>
    </div>
</body>
</html>
```

Code of Conduct:

I __V. Sandeep Reddy(192325096)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *V. Sandeep Reddy*

RUBRICS FOR CALCULATION:

Criteria	Weigh tage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25% 22	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25% 22	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20% 17.6	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20% 17.6	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10% 8.8	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand